AQA 3.2.4 Properties of Period 3 elements and their oxides

AQA A-level Chemistry · Revision questions · Sheet 6 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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19 marks · about 25 minutes

Key information

- $Mg + H_2O(g) \rightarrow MgO + H_2$ (steam); $2Na + 2H_2O \rightarrow 2NaOH + H_2$
- Basic oxides react with acids to give salt + water; acidic oxides react with bases (e.g. $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$)
- Na_2O and MgO are giant ionic and basic: $Na_2O + H_2O \rightarrow 2NaOH$ (pH 12–14); $MgO + H_2O \rightarrow Mg(OH)_2$ (pH 9–10)
- Al_2O_3 is insoluble and amphoteric; SiO_2 is insoluble, giant covalent and acidic (reacts with hot conc NaOH)
- P_4O_{10}, SO_2, SO_3 are simple molecular and acidic: $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$; $SO_2 \rightarrow H_2SO_3$; $SO_3 \rightarrow H_2SO_4$

Section A: Short-answer questions

01 State the type of structure in magnesium oxide and in sulfur dioxide.

[2 marks]

02 Give an equation for the reaction of silicon dioxide with hot, concentrated sodium hydroxide solution.

[1 mark]

03 Give an equation for the reaction of sulfur dioxide with sodium hydroxide solution.

[1 mark]

04 Give an equation for the reaction of aluminium oxide with sodium hydroxide solution to form $NaAl(OH)_4$.

[1 mark]

05 Give an equation for the reaction of sulfur trioxide with water. Suggest the pH of the solution formed.

[2 marks]

Section B: Longer questions

06 The table shows the pH of the solution formed when each Period 3 oxide is added to water.

Oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_3
pH	14	9	7	7	1	0

06.1 Explain why the pH for aluminium oxide and silicon dioxide is 7.

[1 mark]

06.2 Explain the change in pH from Na_2O to SO_3 in terms of the bonding in the oxides.

[2 marks]

06.3 Give an equation for the reaction of magnesium oxide with hydrochloric acid and an equation for the reaction of sulfur dioxide with sodium hydroxide.

[2 marks]

06.4 Explain why magnesium oxide has a higher melting point than sodium oxide.

[2 marks]

07 The melting points of the highest oxides of the elements sodium to sulfur are shown in the table.

Highest oxide of	sodium	magnesium	aluminium	silicon	phosphorus	sulfur
Melting point / K	1548	3125	2345	1883	573	290

07.1 Explain the increase in melting point from sodium oxide to magnesium oxide.

[2 marks]

07.2 Explain why the melting point of the oxide of silicon is much higher than that of the highest oxide of phosphorus.

[3 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	MgO giant ionic (lattice) SO_2 (simple) molecular	*Source: Practice question (written for this set)* Not simple covalent	2
02	$SiO_2 + 2NaOH \rightarrow Na_2SiO_3 + H_2O$	*Source: Practice question (written for this set)*	1
03	$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$	*Source: Practice question (written for this set)* Allow $SO_2 + NaOH \rightarrow NaHSO_3$	1
04	$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$	*Source: Practice question (written for this set)* Allow ionic equation with OH^-	1
05	$SO_3 + H_2O \rightarrow H_2SO_4$ pH 0–1	*Source: Practice question (written for this set)*	2
06.1	They are insoluble in water	*Source: Practice question (written for this set)*	1
06.2	M1 Na_2O / MgO are ionic: O^{2-} reacts with water to form OH^- M2 P_4O_{10} / SO_3 are covalent (molecular) and react with water to form acids that release H^+		2
06.3	$MgO + 2HCl \rightarrow MgCl_2 + H_2O$ $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$		2
06.4	M1 Mg^{2+} has a higher charge / charge density than Na^+ M2 stronger attraction between the ions / stronger ionic bonding		2
07.1	M1 greater charge / charge density on magnesium ion / Mg^{2+} M2 stronger attraction between oppositely charged ions / stronger ionic bonding	*Source: 7405/3 Jun 2018 Q02.3–02.4* Allow magnesium ion is smaller If molecules, IMF or metallic bonding mentioned then CE = 0	2
07.2	M1 (SiO_2) giant covalent / macromolecular M2 (P_4O_{10}) (simple) molecular M3 (covalent) bonds in SiO_2 much stronger than the forces between molecules / IMF in P_4O_{10}	Not simple covalent for M2 M3 dependent on correct M1 and M2	3